electronic jamming, decoy deployment, and target illumination while maintaining autonomous coordination via the Combat Cloud.

Impact: These deployments demonstrated the combat cloud concept, where all aerial assets manned and unmanned, share data in real time, resulting in enhanced situational awareness, mission adaptability, and multi-domain command efficiency. The FCAS model has since become a blueprint for Europe's transition toward sixth-generation cooperative combat ecosystems.

5.8.4 Case study 4: Scientific and environmental applications:

The Tengden TW-356 (2020–present)

In 2020, China's Tengden TW-356 was deployed for high-altitude environmental surveillance over the Qinghai–Tibet Plateau. Its twin-boom HALE design and AI-powered flight management system enabled long-endurance missions exceeding 40 hours. Equipped with hyperspectral imaging sensors, the TW-356 collected data on atmospheric changes, glacial melt patterns, and ecosystem shifts.

Impact: The mission demonstrated the role of fourth-generation UAVs in climate monitoring and ecological assessment, contributing to global environmental databases. It also highlighted how autonomous systems could operate efficiently in remote, high-risk zones where manned flights are impractical.

5.8.5 Case study 5: Defense innovation and indigenous

capability: India's Ghatak UCAV (2022–present)

India's Ghatak UCAV, developed by DRDO–ADE, represents an indigenous leap toward stealth and autonomy. Designed for deep-strike and SEAD (suppression of enemy air defense) missions, the Ghatak's AI-based